
PS Analysis in Data Science: Problem Set 11

Problem 1. Let $U \subset \mathbb{R}^2$ be open and define

$$f(x, y) = x^3 + x - 2y.$$

- (a) Use the Implicit function Theorem to argue that near every point $(a, b) \in \mathbb{R}^2$ with $f(a, b) = 0$, the equation

$$f(x, y) = 0$$

defines y as a continuously differentiable function $y = g(x)$.

- (b) Next, determine g .

Problem 2. Let

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad f(x_1, x_2) = \begin{pmatrix} e^{x_1} \cos x_2 \\ e^{x_1} \sin x_2 \end{pmatrix}.$$

- (a) Compute $f'(x)$.
- (b) Show that $f'(x)$ is invertible for every $x \in \mathbb{R}^2$.
- (c) Use the inverse function theorem to conclude that f is locally invertible near every point $x \in \mathbb{R}^2$.
- (d) Is f globally injective on $\mathbb{R}^2$.

Problem 3. Let

$$M_n = \{x \in \mathbb{R}^d : \|x\|^2 = 1\}.$$

- (a) Show that M_n is a $n - 1$ -dimensional submanifold of $\mathbb{R}^n$.
- (b) Determine the tangent space $T_x(M_n)$ and the normal space $N_x(M_n)$ for all $x \in M_n$.
- (c) Visualize M_2 , $T_x(M_2)$, and $N_x(M_2)$ in Julia for the point $x = (1, 0)^\top$.

Problem 4. Let

$$M = \{x \in \mathbb{R}^3 : g_1(x) = 0, g_2(x) = 0\}, \quad g_1(x) = x_1^2 + x_2^2 - 1, \quad g_2(x) = x_3 - x_1.$$

- (a) Compute the Jacobian $g'(x)$ for $g = (g_1, g_2)^\top$.
- (b) Show that M is a 1-dimensional submanifold of $\mathbb{R}^3$.